

CircuitFLow

Project Data

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1.1 Project Overview

This project, 'CircuitFLow: Smart Water Management, drop by drop aims to significantly reduce domestic water wastage through real-time analysis and usage prediction.

2.1 Team:

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3.1 Deliverables:

1. A functional prototype of the model.
2. A full, working, demonstrable, and replicable hardware working model.
3. Data analytics on extracted data.

4.1 Project Context:

Team Smartopia is developing an optimal water-efficient composite strategy (A domestic water efficiency tool) composed of a low-power sensor, auditory alerting system (hardware), real-time analysis of water-usage data through a local dashboard, and peer-to-peer data communication (software), keeping in view a range of sustainability indicators, such as water-saving potential, cost, and associated energy consumption. The sensor reads the flux of water through domestic taps and transmits the recorded data through hardware connections to a local device, which analyses the usage trend, and predicts future use-patterns. The model aims to sustainably control water-wastage keeping in mind the growing needs

5.1 Current Situation/Existing Works:

Several existing models based on water-conservation have successfully detected inconspicuous water leaks, but have failed to provide lasting solutions owing to a few limitations:

1. They fail to learn from the usage habits of multiple users.
2. They are heavily dependent on the internet and third-party applications for data-analysis and storage, reducing the security of users' data such as location.
3. They implement selective sensing capabilities such as elementary acousic-sensing for water-leaks, which can be erroneous when implemented in non-isolated experiments.
4. They can be oversized and cost ineffective.
5. They do not analyse and tutor their models to the maximum wastage time-periods, thus losing power efficiency.

6.1 Crisis:

Studies performed in Indian cities reveal that the average water consumption was 183 and 215 litres/person/day for standalone households and apartments, respectively. It is estimated that by 2025, 67% of the global population will face moderate to high water stress and half of the population will be suffering constraints in their water supply.

6.2 Factors Influencing Domestic Consumption:

In the context of urban areas in India, two types of buildings, where water consumption takes place, can be considered: residential and non-residential. In residential buildings, water is usually used for flushing, washing clothes and dishes, showering, bathing and satisfying a variety of other uses (such as cooking and drinking). Since the residential sector represents the largest urban water use sector, water saving measures targeting the residential sector could play an important role in reducing the total domestic water consumption.

6.3 Household water-using components:

Breaking consumption into different micro-components of demand has been considered in a number of studies. Several studies have considered micro components such as showers, taps, dishwashers, etc. Bath and shower account for 55% of water-usage in urban households. Efficient showerheads are a prerequisite to water-conservation.

Clothes washing accounts for nearly 20% of per capita household water usage.

7.1 Problem Statement

The global population has increased from 3 to 7 billion people in five decades, placing considerable pressure on water resources. Global and local trends indicate a growing concern over domestic water wastage due to unmonitored and continuous flow from household taps. It is estimated that by 2025, 67% of the global population will face moderate to high water stress and half of the population will be suffering constraints in their water supply. In India, an average household loses nearly 45,500L through tap-leaks annually, aggravating the global crisis. Since the residential sector represents the largest urban water use sector, water saving measures targeting the residential sector could play an important role in reducing the total domestic water consumption.

Users lack real-time awareness about their consumption. Existing models provide cumulative readings, but do not facilitate real-time, user-friendly regulation and analysis. This absence of sustainable and cost-effective user-friendly control mechanisms results in unnecessary water loss, particularly during peak-operation times.

The objective of this project is to develop a smart water usage strategy and system that detects anomalous flow, initiates corrective control actions, and sustainably minimizes domestic water wastage.

By integrating low-power sensors, a flow-control valve(actuator), and a local real-time analytical dashboard, the system aims to promote sustainable usage behaviour.

8.1 Problem Description:

Water wastage from domestic systems remains a significant contributor to urban resource-depletion. Traditional plumbing mechanisms lack automatic regulation, relying entirely on user behaviour to control consumption. Instances such as taps left running unintentionally, pipe leaks, or excessive usage during cleaning activities can lead to substantial losses over time. Most conventional models do not support automated intervention, which would significantly reduce the loss of water. The proposed system addresses these limitations by introducing a hardware-based sensing and regulation solution that measures flowrates through domestic taps, and automatically triggers auditory and shutoff responses, without entirely precluding genuine instances of above-average water use. The collected data is further analysed through novel methods to identify trends, predict future consumption and encourage responsible water management through actionable insights.

The water consumption of a building, which is standardised as “litres/capita/day”, is determined by several factors: climate and weather condition; (2) the types and characteristics of water use micro-components installed in a building; and (3) water use habits. Hence the proposed model seeks to *warn the users* based on their past usage trends, in place of systemically overriding their usage.

9.1 System Architecture

At the high level, the model consists of hardware components as listed below:

a. Arduino UNO MCU

b. Power supply and data extraction via USB-B

c. YF-S401 water-flow sensor

d. Standard jumper wires

e. Resistors

f. HW201 – IR obstacle avoidance sensor module

g. Proportional Solenoid valve

The Arduino MCU operates a voltage of 5V, and possesses 32KB of Flash memory. The microcontroller would run at a clock-speed of 16MHz. The board weighs 25mg, and is powered via a USB-B connection.

The YF-S401 water flow sensor produces a digital output in the working range of 5-18V and has a typical range of 0.3-6L/min. The sensor's accuracy is generally listed as (+/-)5% within the sub-range. To avoid damage and ensure maximum accuracy, this model considers a working range of 0.3L to 3L/min, an optimal operating voltage of 5V and vertical installation at no more than a 5-degree tilt. The sensor is connected to the Arduino via a jumper wire.

The HW201 has an adjustable detection range of 2cm to 30cm and works in the range of 3.3V-5V DC. It produces a digital output, and weighs 3g. It operates optimally at 5V and requires a power supply of at least 20mA. The sensor has a fixed detection angle of 35 degrees. Positioning the sensor within this angle relative to the target is crucial for operation.

9.2 Data and control flow

1. Power and programs are supplied to the MCU via the USB-B cable.
2. The HW201 sensor scans the surroundings for nearby obstacles (hands).
3. If an object (hand) is detected, a voltage signal is sent to the MCU from the sensor. The MCU reads water-flux data from the YF-S401.
4. The valve is initially configured to be shut. When the MCU passes an 'open' instruction, the valve opens to its full capacity.
5. When the valve opens, the built-in timer of the MCU starts.
    a. If the hand is withdrawn before the knob is shut, the HW201 sends a corresponding voltage signal to the MCU.

b. The MCU gradually reduces the aperture of the flowing water-column using the valve, before finally closing the valve completely.

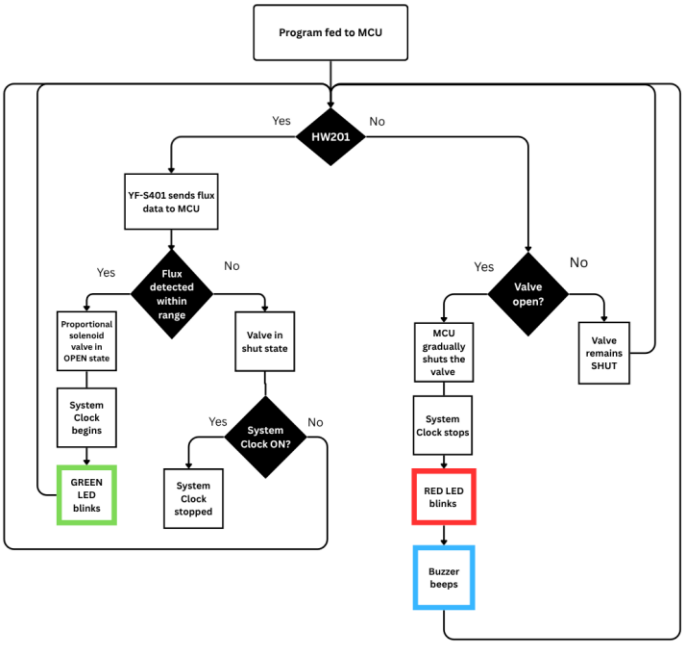
c. The buzzer is activated to produce a sound signal as an indication of the shutting of the valve.

6. If the knob is normally shut after usage,

   a. The valve returns to its original shut state, and the system timer is stopped.

7. The flow-data, timer-data, and status of implementation are extracted from the MCU through the USB-B cable.

8. RED and GREEN LEDs are used to indicate the state of water-flow in the system; A RED LED is used to indicate that no water is flowing through the valve, while a GREEN LED is used to indicate that water is flowing through the valve.



9.3 System Hierarchy

Level	Description
Roles	Suhas: Hardware assembly, management and Arduino
	Ashi: Hardware and Software integration, and Arduino
	Disha: Data extraction, analytics, P2P networking
	Gautam: Data analytics, P2P networking, data extraction.
Records	Members are required to make meaningful commits (logs) after making vital modifications to the system.

10.1 Hardware Design:

1. Block Diagram and schematics:

